

Week 7 Assignment: Exploratory and Explanatory Analysis in Environmental Policy

Aaron Foster

Southern New Hampshire University

CS 540: Data Literacy & Visualization

Prof. Mahmoud Al-Khassaweneh

June 2nd, 2026

Introduction

Environmental policy depends heavily on data. Decisions about emissions standards, water quality regulations, conservation funding, and climate adaptation all rely on analysts who can both investigate complex datasets and communicate findings to people who shape law. Two complementary modes of analysis support this work: exploratory analysis, which helps researchers understand what is happening in the data, and explanatory analysis, which helps them convey those findings to decision-makers. Understanding the difference between the two is essential for anyone producing data-driven work in this field.

Defining Exploratory and Explanatory Analysis

Exploratory analysis is the open-ended process of examining a dataset to understand its structure, surface patterns, and generate questions. The analyst usually does not know what they are looking for at the start. In environmental policy, an exploratory analysis might involve pulling ten years of air quality readings from monitoring stations across a state and looking for unexpected spikes, seasonal patterns, or geographic clusters of poor air quality. The goal is discovery, not conclusion.

Unlike exploration, explanatory analysis happens after you've discovered what the data means. It's the art of selecting, framing, and sharing specific insights so a targeted audience can understand and act on them. For instance, an environmental policy analyst might present a one-slide summary to state legislators proving that local air pollution breached federal limits on

47 days, using that data to argue for stricter industrial permits. The ultimate objective is clarity and persuasion.

Key Differences Between the Two Approaches

The two approaches differ in several important ways. First, the purpose differs: exploratory work is about discovery, while explanatory work is about communication. Second, the audience differs. Exploratory analysis is usually done by the analyst for the analyst, or for a small technical team, while explanatory analysis is built for an external audience such as policymakers, the public, or agency leadership. Third, the volume and polish of output differ. Exploratory analysis often produces dozens of rough charts, many of which lead nowhere, whereas explanatory analysis typically distills the work down to one or two carefully designed visuals. Fourth, the questions differ. Exploratory analysis asks "what is in this data?" while explanatory analysis answers "what does this data mean for the decision in front of us?" Finally, the tolerance for messiness differs. Exploratory charts can be cluttered and unlabeled because the analyst understands them; explanatory charts must be clean, annotated, and self-explanatory.

Where Each Fits in the Data Analysis Process

Exploratory analysis occurs early in the data pipeline (following collection and cleaning) and focuses on data profiling and hypothesis generation. For example, an environmental analyst might spend hours studying new deforestation satellite imagery during this stage before making any assumptions. Explanatory analysis, by contrast, occurs at the end of the workflow. It belongs to the communication phase, where validated findings become recommendations for

stakeholders. Ultimately, exploration informs the analyst, while explanation informs the decision-maker.

Contribution to Decision Making in Environmental Policy

Both analytical approaches impact environmental policy in unique ways. Exploratory analysis uncovers hidden issues before they escalate—such as revealing an unnoticed link between a specific pollutant and local pediatric asthma rates. Without this initial groundwork, regulatory agencies are trapped in a purely reactive cycle. Conversely, explanatory analysis is the catalyst for policy change. Because lawmakers lack the time to interpret raw data, they rely on clear summaries that highlight the evidence and recommend solutions. For instance, a well-crafted explanatory chart detailing projected carbon emission cuts can decide the fate of a proposed regulation. Ultimately, they form a vital pipeline: exploration uncovers the problem, and explanation drives the solution.

Improving Professional Work

Understanding the distinction sharpens an analyst's professional practice in several ways. It clarifies which tools and techniques to reach for at each stage—for example, using flexible scripting and interactive dashboards during exploration but switching to polished static visuals for stakeholder briefings. It also helps the analyst manage their time, since over-polishing exploratory charts is wasted effort, while under-polishing explanatory ones costs credibility. Most importantly, it forces the analyst to think carefully about audience. A chart that helps a data scientist understand a dataset is rarely the same chart that helps a city council understand a policy

choice. Recognizing that difference is a hallmark of effective data work.

Visualization Examples

An exploratory visualization of air quality data might feature a dense, small-multiples grid displaying five-year trends for every monitoring station in a state. This technical layout allows an analyst to scan for anomalies, seasonality, and regional patterns. In contrast, an explanatory visualization of the same dataset strips away the noise, offering a single annotated line chart of one specific city. By marking exactly when a new emissions law was enacted and adding a text callout of the resulting air quality improvement, this clean, focused graphic is designed to deliver a single, compelling takeaway to non-technical stakeholders.

Conclusion

Exploratory and explanatory analysis are not competing approaches but complementary stages of the same workflow. In environmental policy, where evidence must travel from raw monitoring data to legislative action, both are indispensable. Exploration uncovers the problems worth solving, and explanation makes the case for solving them. An analyst who understands when to use each, and who can move smoothly between the two, is far more valuable to the teams and decisions they support.